Part 1

- First, we will install Flask app directly without docker:
  → python3, pip3 are installed
  →and getting the source code from GitHub by git clone

```
aiysha@aiysha-VMware-Virtual-Platform:~$ python3 --version
pip3 --version
Python 3.13.7
pip 25.1.1 from /usr/lib/python3/dist-packages/pip (python 3.13)
aiysha@aiysha-VMware-Virtual-Platform:~$ git clone https://github.com/meldafrawi/basic-flask-app.git
```

- When then trying to install the requirements.txt
  or the even trying manual installation by pip3 install flask
  this error occurs:

```
aiysha@aiysha-VMware-Virtual-Platform:~/basic-flask-app$ pip3 install -r requirements.txt
error: externally-managed-environment

× This environment is externally managed
╰─> To install Python packages system-wide, try apt install
    python3-xyz, where xyz is the package you are trying to
    install.

    If you wish to install a non-Debian-packaged Python package,
    create a virtual environment using python3 -m venv path/to/venv.
    Then use path/to/venv/bin/python and path/to/venv/bin/pip. Make
    sure you have python3-full installed.

    If you wish to install a non-Debian packaged Python application,
    it may be easiest to use pipx install xyz, which will manage a
    virtual environment for you. Make sure you have pipx installed.
```

  → why this error occurs?
  Newer versions of Python on Ubuntu (like Python 3.12/3.13) follow **PEP 668**, which prevents pip from installing packages directly into the system Python.
  This is to avoid breaking system tools that depend on Python.

- Better way? To use virtual environment:
  → installing python3-venv
  → creating a new virtual environment named venv by python3 –m venv venv
  →so, now /venv appears containing its own copy of python and pip
  →activating the virtual environment by source venv/bin/activate

  now, Any pip install or python command applies only to this environment, not the system.
  So here the venv acts as a sandbox for python projects.

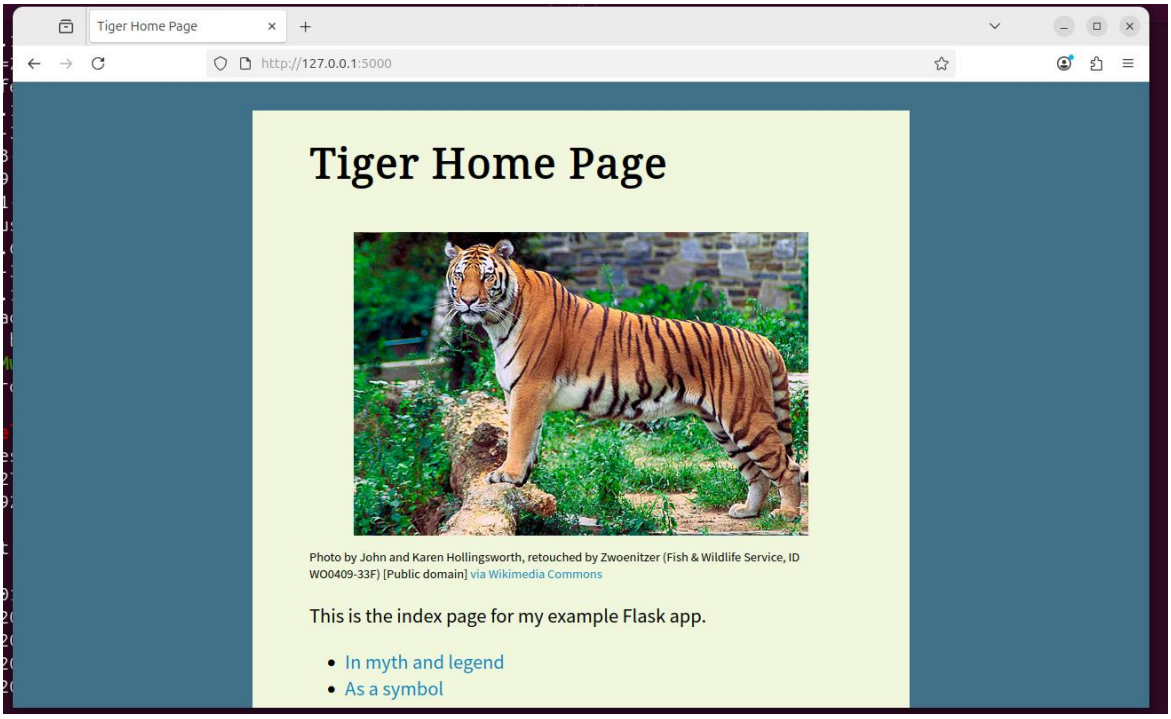
```
aiysha@aiysha-VMware-Virtual-Platform:~/basic-flask-app$ sudo apt install python3-venv -y
python3 -m venv venv
source venv/bin/activate
Installing:
  python3-venv

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 111
  Download size: 1,028 B
  Space needed: 6,144 B / 8,856 MB available

Get:1 http://eg.archive.ubuntu.com/ubuntu questing/universe amd64 python3-venv amd64 3.13.7-1 [1,028 B]
Fetched 1,028 B in 0s (3,952 B/s)
Selecting previously unselected package python3-venv.
(Reading database ... 148751 files and directories currently installed.)
Preparing to unpack .../python3-venv_3.13.7-1_amd64.deb ...
Unpacking python3-venv (3.13.7-1) ...
Setting up python3-venv (3.13.7-1) ...
(venv) aiysha@aiysha-VMware-Virtual-Platform:~/basic-flask-app$ pip install flask
Collecting flask
  Downloading flask-3.1.3-py3-none-any.whl.metadata (3.2 kB)
Collecting blinker>=1.9.0 (from flask)
  Downloading blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Collecting click>=8.1.3 (from flask)
  Downloading click-8.3.1-py3-none-any.whl.metadata (2.6 kB)
Collecting itsdangerous>=2.2.0 (from flask)
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting jinja2>=3.1.2 (from flask)
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
```

- Another solutions:|
  →**Downgrading Python** or forcing system installs are workarounds, but they're fragile and can cause problems later.
  →**override the restriction** with:
  pip install flask requests --break-system-packages
  But this risks breaking Ubuntu's Python environment. If you install conflicting versions, system tools may stop working. That's why it's discouraged.

- Run Flask:

```
(venv) aiysha@aiysha-VMware-Virtual-Platform:~/basic-flask-app$ python routes.py
 * Serving Flask app 'routes'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://192.168.164.129:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 368-101-128
127.0.0.1 - - [01/Mar/2026 23:07:04] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [01/Mar/2026 23:07:04] "GET /static/css/main.css HTTP/1.1" 200 -
127.0.0.1 - - [01/Mar/2026 23:07:04] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [01/Mar/2026 23:07:04] "GET /static/images/tiger.jpg HTTP/1.1" 200 -
S
```

- Opening the address in the browser:



- Second, building a Flask Docker image:

  → Dockerfile:

```
aiysha@aiysha-VMware-Virtual-Platform: ~/basic-flask-app — vim Dockerfile
~/basic-flask-app

FROM python:3.12-alpine

WORKDIR /app

COPY . /app

RUN pip install flask requests

ENTRYPOINT ["python"]

CMD ["routes.py"]
```

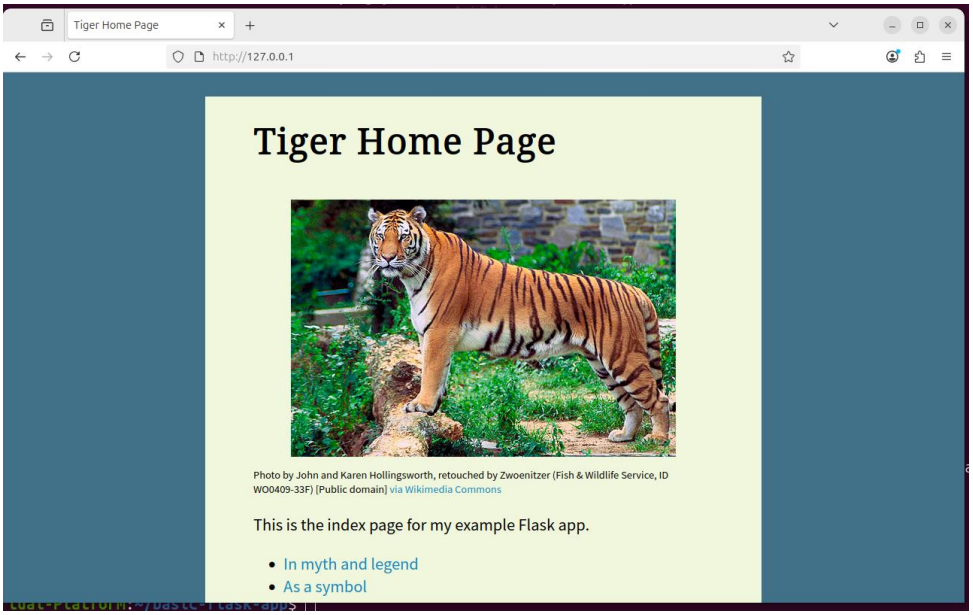
- Building image:

```
aiysha@aiysha-VMware-Virtual-Platform:~/basic-flask-app$ sudo docker build -t iti-flask-lab2 .
[+] Building 18.3s (9/9) FINISHED                                docker:default
 => [internal] load build definition from Dockerfile                          0.0s
 => => transferring dockerfile: 164B                                          0.0s
 => [internal] load metadata for docker.io/library/python:3.12-alp            2.7s
 => [internal] load .dockerignore                                             0.0s
 => => transferring context: 2B                                               0.0s
 => [1/4] FROM docker.io/library/python:3.12-alpine@sha256:2d91681            5.1s
 => => resolve docker.io/library/python:3.12-alpine@sha256:2d91681            0.0s
 => => sha256:77e59d578b52ba7be2dcd4414d9ef235af6ea92f 247B / 247B            0.5s
 => => sha256:aca07fe076f12b62834e15448c6ec185 460.94kB / 460.94kB            0.9s
 => => sha256:4e18f69996e78732b37aff52d0d056f151 13.74MB / 13.74MB            4.6s
 => => extracting sha256:aca07fe076f12b62834e15448c6ec1858845fe802            0.1s
 => => extracting sha256:4e18f69996e78732b37aff52d0d056f1513dfe14a            0.4s
 => => extracting sha256:77e59d578b52ba7be2dcd4414d9ef235af6ea92f5            0.0s
 => [internal] load build context                                             0.5s
 => => transferring context: 15.68MB                                          0.4s
 => [2/4] WORKDIR /app                                                        0.2s
 => [3/4] COPY . /app                                                         0.2s
 => [4/4] RUN pip install flask requests                                      8.7s
 => exporting to image                                                        1.3s
 => => exporting layers                                                       0.8s
 => => exporting manifest sha256:6215e799eb1192a1385ca1fe3b3a453c0            0.0s
 => => exporting config sha256:6378921561036fc90827b42a8f4e77312ac            0.0s
 => => exporting attestation manifest sha256:311ff856470b36690e4a0            0.0s
 => => exporting manifest list sha256:8a4bb26897e3b484c3611f6ea45f            0.0s
 => => naming to docker.io/library/iti-flask-lab2:latest                      0.0s
 => => unpacking to docker.io/library/iti-flask-lab2:latest                   0.4s
aiysha@aiysha-VMware-Virtual-Platform:~/basic-flask-app$
```

- Run:

```
aiysha@aiysha-VMware-Virtual-Platform:~/basic-flask-app$ sudo docker run -d --name flask-lab2 --memory=100m -p 80:5000 iti-flask-lab2
c2a2a3f66a8a4774f5d1aa636692f38b04715fb801a4780624aa61677273186a
```

- Verification it worked:



- Third, Push to docker hub:

```
iysha@aiysha-VMware-Virtual-Platform:~/basic-flask-app$ sudo docker login -u aiy4a

 Info → A Personal Access Token (PAT) can be used instead.
        To create a PAT, visit https://app.docker.com/settings


assword:

ARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
onfigure a credential helper to remove this warning. See
ttps://docs.docker.com/go/credential-store/

ogin Succeeded
iysha@aiysha-VMware-Virtual-Platform:~/basic-flask-app$ sudo docker tag iti-flask-lab2 aiy4a/docker-iti:latest
iysha@aiysha-VMware-Virtual-Platform:~/basic-flask-app$ sudo docker push aiy4a/docker-iti:latest
he push refers to repository [docker.io/aiy4a/docker-iti]
7e59d578b52: Pushed
e18f69996e7: Pushed
6f7458f1b62: Pushed
448ccacfff0: Pushed
b15a3df846f: Pushed
0ee021eb3dc: Pushed
89002ba0eae: Mounted from library/nginx
ca07fe076f1: Pushed
atest: digest: sha256:8a4bb26897e3b484c3611f6ea45f50843dbb6fb67138a915cfcff2c7017a835c size: 856
iysha@aiysha-VMware-Virtual-Platform:~/basic-flask-app$
```

Part 2

- Create the network:

```
aiysha@aiysha-VMware-Virtual-Platform:~$ sudo docker network create --driver bridge --subnet 10.0.0.0/8 iti-network
[sudo: authenticate] Password:
b72759634e72550b3d7748a5ed9202ff527eb00bf4615e61ae0a0d5fdf26e3c1
```

- Create a directory on the host server so further we could mount bind the html file:

```
aiysha@aiysha-VMware-Virtual-Platform:~$ mkdir ~/iti-web
```

- Inside the directory, creating index.html:

```
aiysha@aiysha-VMware-Virtual-Platform:~/iti-web$ cat index.html
<h1>Lab 2 ITI - Aiysha</h1>
```

- Running the container:

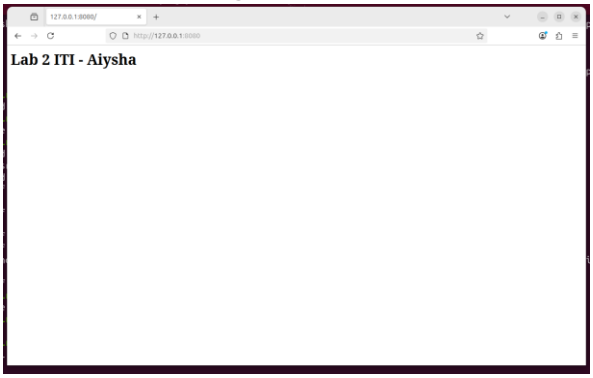
```
aiysha@aiysha-VMware-Virtual-Platform:~$ sudo docker run -d --name webserver-iti --network iti-network -p 8080:8080  -v ~/iti-web:/usr/local/apache2/htdo
cs:ro  httpd:alpine
e2f4a16e224c07acaf632db83dbf3de3dbe1d8ed23d5a5ed69e5ec2e96a64374
```

- Inside the container, change the listening port from 80 to 080:

```
aiysha@aiysha-VMware-Virtual-Platform:~/iti-web$ sudo docker exec -it webserver-iti sh
/usr/local/apache2 # vi /usr/local/apache2/conf/httpd.conf
/usr/local/apache2 #
```

```
#
# Change this to Listen on specific IP addresses as shown below to
# prevent Apache from glomming onto all bound IP addresses.
#
#Listen 12.34.56.78:80
Listen 8080
```

- Restarting the container to validate changes:

```
aiysha@aiysha-VMware-Virtual-Platform:~/iti-web$ sudo docker restart webserver-iti
webserver-iti
```

- Service working: